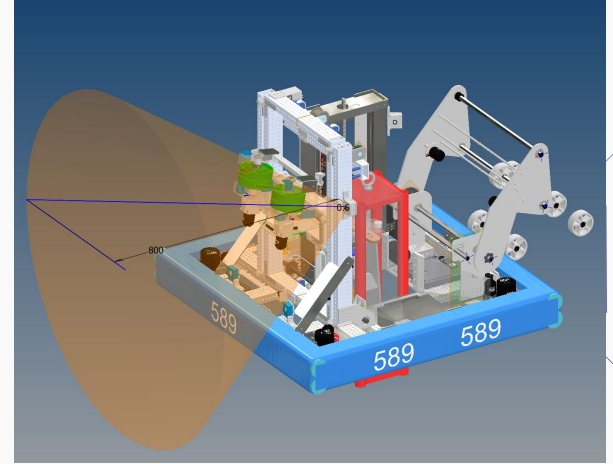
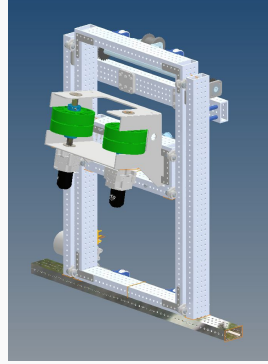


FRC 589 Robot Design



Vela Benedicto

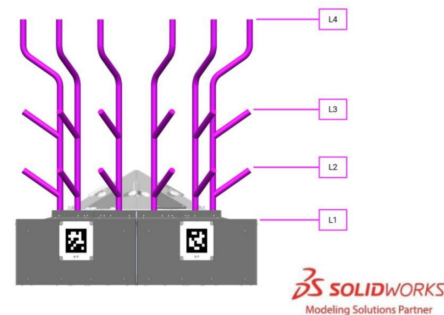
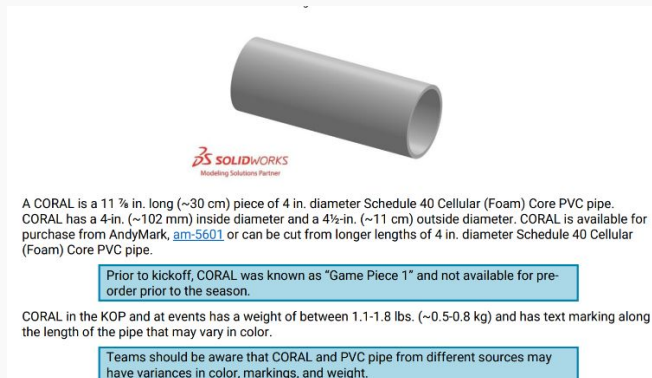


COMPETITION FORMAT

- FRC (FIRST Robotics Competition): **6-8 weeks** to build a 120-lb robot from scratch to complete certain elements of a yearly-released game
- **Goal: accrue as many points as possible w/in a 2min 30 sec period**

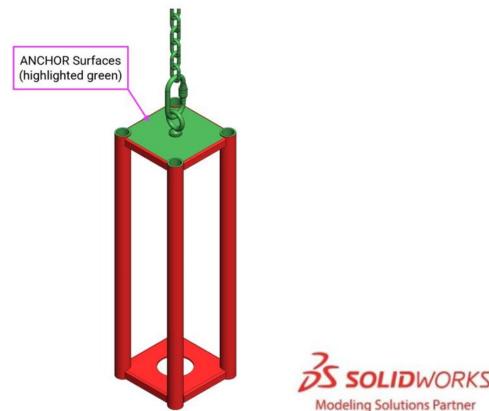
2025 OBJECTIVES

- Place coral (hollow PVC cylinder) to rungs on the reef (pink → right)
- Hang from the cage for endgame



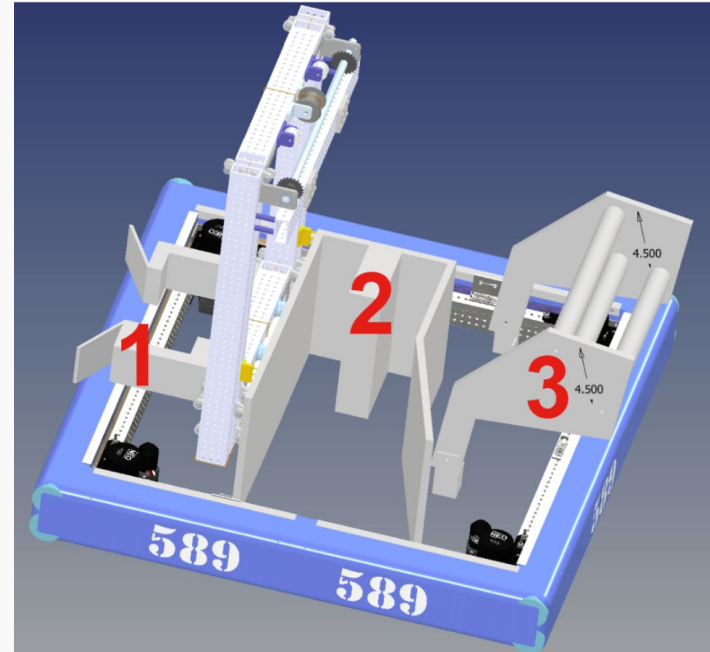
The base of each REEF has a trough (L1) into which ROBOTS can score CORAL. The trough includes the angled and vertical surfaces, as well as the top of the front edge as shown in [Figure 5-7](#). The front edge of the trough is 1 ft. 6 in. (~46 cm) off the carpet.

Figure 5-11: ANCHOR Surfaces



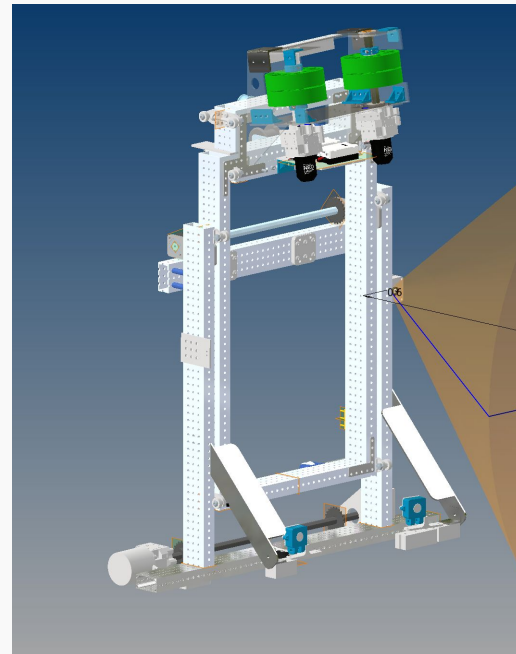
ROBOT OVERVIEW

- 1.** Coral station intake + “shoot” into L2/L3
- 2.** Deep cage climber
- 3.** Ground intake for coral + deposit into trough (L1)



THOUGHT PROCESS

1. “WRIST” GOALS
 - a. No motor for carriage → too heavy
 - b. Intake from a human player (at coral station)
 - c. Shoot L2/L3 consistently → same angle/points are optimal
 - d. Sustain coral hold when motors are off
2. ELEVATOR GOALS
 - a. 2nd stage
 - b. Needs to reach X amt of height
 - c. Min 0th stage width → need space for other components
3. QUESTIONS
 - a. How tall can carriage be?
 - b. If carriage is static, is 6.21” secure enough to hold game piece?
 - c. Elevator mounting space
 - d. How to mount
 - e. Motor placement for wrist



DESIGN PROCESS

1. Sketch ideas in engineering journal
2. Perform spreadsheet calculations
3. Google/YT arm/wrist ideas →
screenshotted/hyperlinked into XMind (mind map)
4. Make *a pros/cons list* for every idea...
5. Perform preliminary prototyping
6. CAD! :D
7. Build XD

SPREADSHEET :D CALCULATIONS

Gearbox

Empirical Specifications			
A	B	C	D
Motor	Nominal Voltage	Motor Kv (Kv)	Free Speed / Free Running C Stall Current (A) Stall Torque (N) Peak Output Power (W)
NEO Brushless	12	473	5676 1.8 105 2.5 406

Variables		Imperial	Metric
W	5 lb	3.62873896 kg	
W (Motor)	5 lb	2.26796185 kg	
W (Motor)	5 lb	2.26796185 kg	
W (Motor)	18.875 in	0.479425 m	
W (Motor)	1 in	0.0254 m	
W (Motor)	1 in	20 A	
W (Motor)	1 in	0.4 Nm	

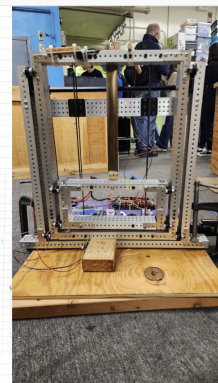
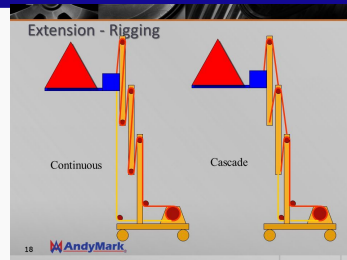
Derived	Exact	Approximate
Fixed	102.2397202 N	
Constant	0.166952068 m	
Approximate	3.004049551 rev	
Approximate	180.2429731 rpm	
Approximate	2.596888893 Nm	

Name	Contents	Price
REV MAXPlanetary Gearbox	gearbox front and back, 1/2" output; hardware; no modify NEO attachment; 1/2" hex shaft (3in)	
Base Kit	MAXPlanetary 9:1 Cartridge Total:	\$36.00
1strev	13.71106372 A	
1strev < 1strev	Yes -)	
1strev	3765.668621 rpm	
1strev	418.4078246 rpm	
1strev > 1strev	Yes -)	
VersaPlanetary Gearbox	gearbox front and back; ring gear and 9:1 gear; hardware; CIM motor mount adapter; universal female output universal female to 1/2" https://www.vexrobotics.com/versaplanetary/box-adapters/ Requires NEO 1" shaft to be out	\$39.95
1strev	1.030969512 Nm	
1strev	3765.668621 rpm	
1strev	0.4307831943 A	

$$\omega_{motor} = .75 * \left(-\frac{\omega_{free}}{\tau_{stall}} * \tau + \omega_{free} \right) =$$

$$\omega_{motor} = .75 * \left(-\frac{\omega_{free}}{\tau_{stall}} * \tau + \omega_{free} \right) =$$

Elevator travel distance

[illegible]

More SS calculations!

Deep Cage Climb Calculations

Saturday, January 18, 2025 10:37 PM

$$\omega = \frac{v}{r} \text{ (angular speed)}$$

$$v = r\omega$$

$$140 \left(\frac{1}{3} \right) = 140 \left(\frac{1}{3} \right) \cdot \frac{1}{3} \cdot \frac{1}{3}$$

$$\tau = 140 \left(\frac{1}{3} \right) \left(\frac{1}{3} \right)$$

$$\omega = v(4.7)$$

$$= 2.37 \frac{\text{rad}}{\text{s}}$$

$$\frac{140}{3} \cdot \frac{1}{3} \cdot \frac{1}{3} \cdot \frac{1}{3}$$

$$\tau = Fv$$

$$\tau = \frac{Fv}{\omega}$$

$$\tau = \frac{140 \left(\frac{8 \text{ in}}{3} \right)}{\frac{32\pi \text{ rad}}{5}}$$

$$\tau = \frac{140 \left(\frac{8 \text{ in}}{3} \right)}{32\pi \text{ rad}}$$

$$\tau = \frac{11.66 \text{ lb in}}{\pi \text{ rad}}$$

$$\omega = v \left(\frac{1 \text{ rev}}{0.5 \text{ in}} \right) \left(\frac{2\pi \text{ rad}}{1 \text{ rev}} \right)$$

$$\omega = v \left(\frac{4\pi \text{ rad}}{\text{in}} \right)$$

$$\omega = \frac{8 \text{ in}}{3 \text{ s}} \left(\frac{4\pi \text{ rad}}{\text{in}} \right)$$

$$\omega = \frac{32\pi \text{ rad}}{5}$$

1) determine angular speed $\left(\frac{\text{rad}}{\text{s}} \right)$
(how fast lead screw must spin to move 8 in/3s - v)

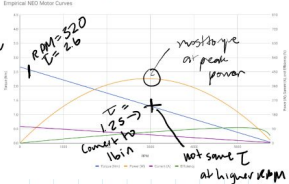
Assume F = robot weight

2) solve for τ (input for lead screw)

3) convert torque to in lb (needs req)

→ want 320 RPM

→ use 10:1 gear ratio RPM



Power

↓ as RPM ↑
bec loads EMF

Units	Conversion	Units
Weight	145 lb	
Travel Distance	8 in	
Time	3 s	
Velocity	2.66666667 in/s	
Angular Speed	33.51032164 rad/s	
Pitch	0.125 in	
Starts	4 Starts	
Lead	0.5 in	
Torque	11.53873337 in lb	

$$\frac{2.66 \text{ in/s} \cdot 2\pi \text{ rad}}{0.5 \text{ in}} = \frac{\text{rad}}{\text{s}}$$

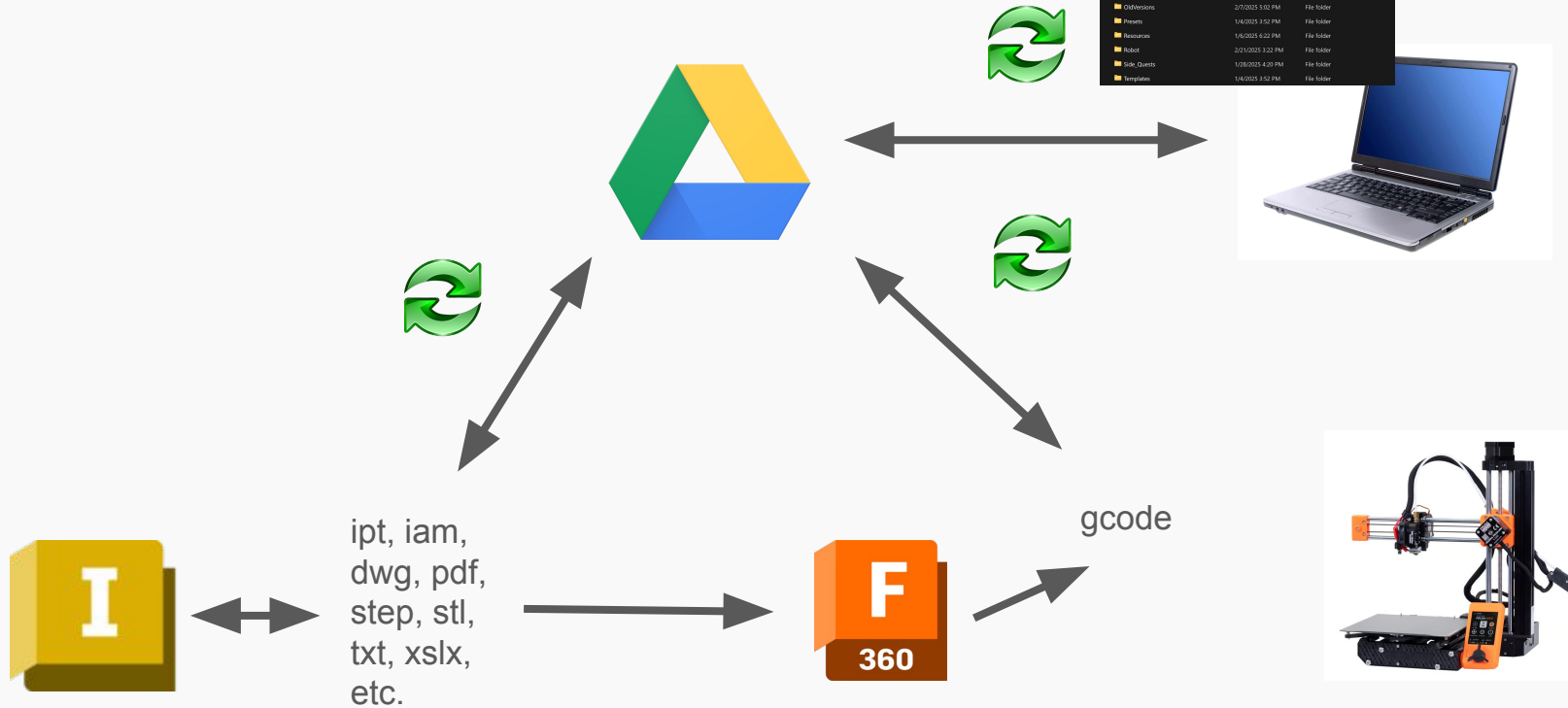
linear → angular speed

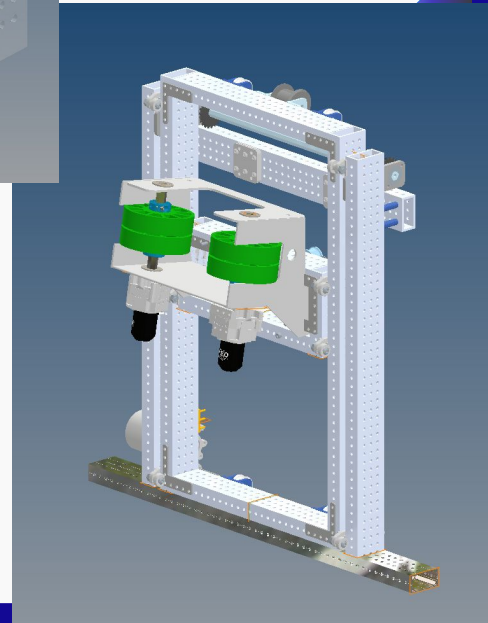
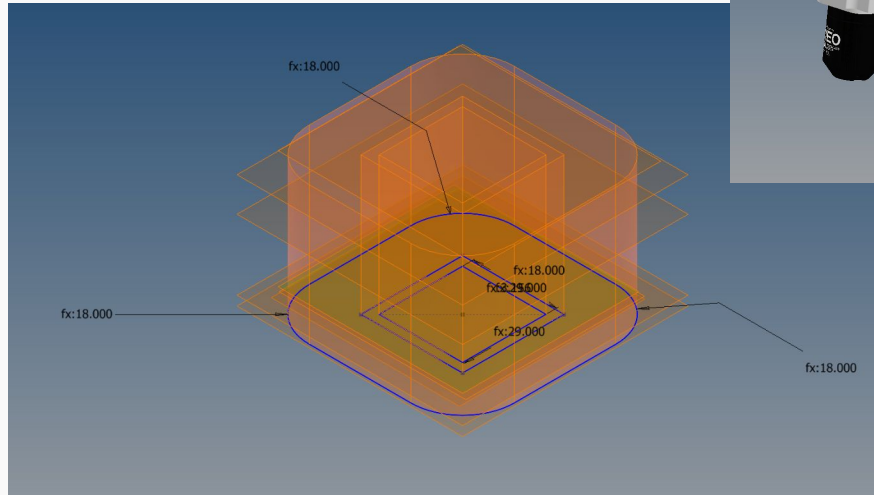
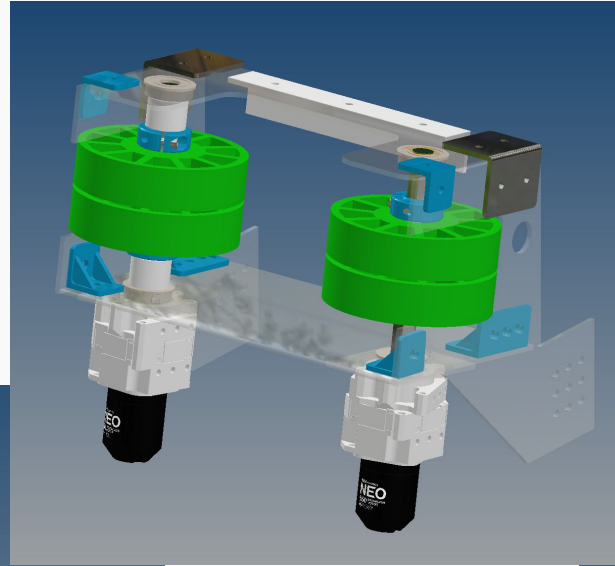
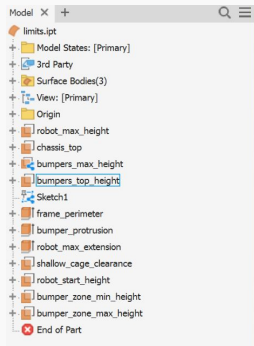
$$\tau = \frac{145 \text{ lb} \cdot 2.66 \text{ in/s}}{\text{rad/s}} = \text{in lb}$$

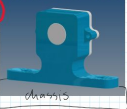
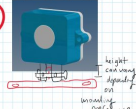
$$\frac{\text{rad}}{\text{s}} \cdot \frac{1 \text{ rev}}{2\pi} \cdot \frac{60 \text{ s}}{1 \text{ min}}$$

$$\frac{33 \text{ rad}}{\text{s}} \cdot \frac{30}{\pi}$$

WORKFLOW





	①	②	③
			
pros	- just one 3D-printed piece	- can be mounted on elevator bumper 1" more along flange already there (requires 15" extension) - less 3D print height -> more reliable	- can be mounted without 1" more along on top of chassis - easily removable (only need to undo bolt for electronics to wire)
Cons	- full 3D print height (~1.5") to see above bumper - not sure if easily removable (if riveted, may damage 3D print?)	- need to move bumper quick release - not sure if easily removable (if riveted, may damage 3D print?)	- potentially more complicated w/ bolt? (additional stresses, maybe spacers) tool access?

	4" carriage	5" carriage
pros	- simpler polycarb mounting - keep elevator travel - more tolerance for robot perimeter (~1/2")	- 3" ID - easier to mount side plates on elevator (see diagram) - can protect motor better (?) - depends on extra polycarb angle
Cons	- 2" ID - harder to mount side plates (...) - motor needs 3D printed protection (managable, just extra)	- more complicated polycarb mounting (pieces) - narrow margin of error from robot perimeter (only ~0.3") - lose 2" of elevator travel
	